

Predicting Customer Churn in Subscription Services Using Interpretable Machine Learning

A Business Decision-Support Framework for Retention Management

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Abstract—Customer churn erodes the recurring revenue on which subscription businesses in telecommunications, banking, and electronic commerce depend, yet the machine learning models that predict churn most accurately are often the ones that business users understand least. This paper proposes a framework that treats predictive accuracy and interpretability as joint design objectives rather than competing ones. Three classifiers of increasing complexity, namely logistic regression, random forest, and gradient-boosted trees, are trained and compared on three public datasets drawn from the telecommunications, banking, and electronic commerce sectors, allowing model behaviour to be examined across domains rather than within a single benchmark. Post-hoc explanation methods based on Shapley values and local surrogate models are applied to every prediction, and their outputs are mapped through a retention intervention taxonomy that converts statistical churn drivers into prioritised, customer-specific retention actions. A dashboard designed for non-technical customer relationship management staff presents each at-risk customer together with the three strongest drivers of the risk estimate and a recommended intervention. Models are assessed with precision, recall, F1-score, area under the receiver operating characteristic curve, Matthews correlation coefficient, and probability calibration, alongside a structured assessment of explanation quality. The framework addresses a persistent gap between churn prediction research, which has concentrated on accuracy, and retention practice, which requires reasons and recommended actions. Ethical safeguards covering data privacy, fairness, and human oversight are built into the design rather than appended to it.

Keywords—customer churn; interpretable machine learning; explainable artificial intelligence; decision-support systems; customer retention; Shapley values

I. INTRODUCTION

Subscription businesses live or die by their ability to keep the customers they already have. Customer churn, understood here as the voluntary decision of a subscriber to terminate a service relationship, is among the most closely watched indicators in telecommunications, banking, software-as-a-service, and electronic commerce, because the economics of retention and acquisition are sharply asymmetric. Acquiring a new customer typically costs several times more than retaining an existing one, and modest improvements in retention compound into disproportionately large gains in customer lifetime value [1], [2]. Lemmens and Gupta estimate that firms which target retention campaigns using loss-aware models can

improve campaign profit substantially relative to conventional targeting, which suggests that the value of churn modelling lies less in the prediction itself than in the decisions the prediction informs [3].

Machine learning (ML) has become the standard technical response to the churn problem. Comparative studies report that ensemble methods, particularly random forests and gradient-boosted decision trees, outperform simpler classifiers on the benchmark datasets used in this literature [4], [5]. Yet accuracy gains of this kind have not translated into equally widespread operational adoption. A retention manager who is told only that a customer has a churn probability of 0.82 has no basis for choosing between a price concession, a service upgrade, or a support call, and no way to judge whether the model's reasoning is sound. Rudin has argued forcefully that opaque models are routinely deployed in settings where the cost of an unexplained error is high, and that the absence of interpretability is a deployment risk rather than a cosmetic shortcoming [6]. In regulated sectors the concern is sharper still, since European data protection rules give individuals a qualified right to meaningful information about automated decisions that affect them [7].

Explainable artificial intelligence (XAI) offers a route out of this impasse. Post-hoc attribution methods such as SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) attach a quantified contribution to every input feature of every individual prediction [8], [9]. A growing number of churn studies now include such explanations [10]. What these studies generally do not include is the final step that retention practice requires: a systematic translation of explanation outputs into concrete retention actions that a non-technical customer relationship management (CRM) team can execute and audit. Explanations are produced for the data scientist and stop at the data scientist.

This paper proposes an interpretable churn prediction framework in which the explanation layer is designed from the outset as the input to a business decision-support system rather than as a diagnostic afterthought. The proposed work makes five contributions.

- A comparative evaluation of logistic regression, random forest, and XGBoost on three public datasets spanning telecommunications, banking, and electronic commerce, so that conclusions about model behaviour do not rest on a single domain.

- A dual-axis assessment protocol that scores models on discrimination and calibration as well as on the stability and fidelity of the explanations they support.
- An explanation-to-action taxonomy that maps SHAP-identified churn drivers onto a structured set of retention interventions with assigned business owners.
- A decision-support dashboard, aimed at non-technical CRM users, that presents churn probability, the leading drivers of that probability, and a recommended intervention for each customer, with batch scoring of uploaded customer lists.
- An explicit treatment of the ethical obligations that follow from using customer data to drive differential treatment, covering privacy, fairness, and human oversight.

The remainder of the paper is organised as follows. Section II reviews prior work on churn prediction, explainable ML, and decision support, and states the research gap. Section III presents the proposed framework. Section IV details the methodology, including datasets, preparation, feature engineering, model selection, and evaluation. Section V develops the explainability and decision-support components. Section VI addresses ethical and practical considerations. Section VII discusses limitations and future work, and Section VIII concludes.

II. RELATED WORK

A. Machine Learning for Churn Prediction

Churn prediction has a long history in analytical CRM. Early surveys already framed it as a classification problem over behavioural and contractual attributes and noted the difficulty of moving from prediction to management action [1]. Verbeke et al. broadened the evaluation question by showing that the model which maximises statistical accuracy is not necessarily the model that maximises campaign profit, an observation that anticipates much of the present argument about the insufficiency of accuracy-centric evaluation [2]. Vafeiadis et al. benchmarked a set of standard classifiers on telecommunications data and found boosted ensembles superior to single models, a result replicated many times since [4]. Ahmad et al. reached similar conclusions at industrial scale, training tree ensembles on operator data covering millions of subscribers and reporting area under the receiver operating characteristic curve values above 0.93 [5].

Two methodological strands within this literature matter for the present work. The first concerns model structure. De Caigny et al. proposed a hybrid of logistic regression and decision trees precisely because end users found pure ensembles unintelligible, showing that the interpretability concern has been visible inside the churn literature for some time [11]. The second concerns features. Alboukaey et al. demonstrated that dynamic behavioural features computed over daily windows outperform static monthly aggregates in mobile telecommunications, which motivates the inclusion of recency and frequency features in the present design [12]. Class imbalance is a recurring obstacle across the strand: churners are usually a small minority of records, and Burez and Van den

Poel showed that sampling strategy affects reported performance as much as classifier choice does [13]. The synthetic minority over-sampling technique (SMOTE) of Chawla et al. remains the most common remedy [14].

A recent systematic review by Imani et al., covering 240 studies published between 2020 and 2024, offers a useful summary of where the field now stands: gradient boosting dominates on tabular churn data, deep learning delivers at best marginal gains outside sequential or textual inputs, and the great majority of studies still evaluate on a single dataset from a single sector [15]. The review is also candid about what the literature neglects, listing interpretability, class imbalance handling, and deployment realism among the most persistent gaps. Read critically, the accumulated evidence supports a clear conclusion. The accuracy question is largely settled, and the open problems are comparative generalisation across domains and the usability of model outputs, which are exactly the problems this project takes up.

B. Explainable Artificial Intelligence

XAI research divides into intrinsically interpretable models, whose structure can be read directly, and post-hoc methods, which explain a trained model from the outside [16], [17]. Among post-hoc methods, two dominate applied work. LIME fits a sparse linear surrogate to the model's behaviour in the neighbourhood of a single instance, producing a local explanation that is intuitive but sensitive to the sampling used to build the neighbourhood [9]. SHAP grounds attribution in cooperative game theory, assigning each feature its Shapley value, and is unique in satisfying local accuracy, missingness, and consistency simultaneously [8]. For tree ensembles, the TreeExplainer algorithm computes exact Shapley values in polynomial time, which makes SHAP practical for the model classes that dominate churn prediction [18].

The two methods are complementary rather than interchangeable. SHAP offers axiomatic guarantees and consistent global summaries but assumes feature independence in its common approximations, which can distribute credit misleadingly across correlated features. LIME is model-agnostic and cheap for single instances but is known to produce unstable explanations when the perturbation process is re-run. Rudin's critique cuts deeper: post-hoc explanations describe the model, not the world, and can lend unwarranted authority to a flawed predictor [6]. The design consequence adopted in this project is threefold. A transparent baseline model is retained throughout as a reference point, SHAP explanations are cross-checked against LIME on a common sample, and explanations are audited against domain knowledge before being exposed to business users. Recent applied studies, such as the multi-model ensemble analysis of El Attar and El-Hajj, show SHAP being used to rank churn drivers on telecommunications data, but the interpretation is offered to the reader as evidence of model plausibility rather than operationalised for retention staff [10].

C. Decision Support and Profit-Aware Retention

A separate research strand asks what a churn model is for. Verbraken et al. formalised the expected maximum profit criterion, which evaluates a classifier by the profit of the retention campaign it induces rather than by its error rate [19].

Höppner et al. pushed the same logic into model training by deriving profit-driven decision trees whose split criterion is monetary rather than statistical [20]. Lemmens and Gupta showed empirically that targeting the customers whose retention creates the most value differs from targeting those most likely to churn [3]. Uplift modelling generalises the point by estimating the causal effect of an intervention on each customer, so that campaigns are aimed where they change behaviour rather than where risk is merely high [21].

This strand supplies the economic rationale for a decision-support layer, but it has developed almost independently of the XAI strand. Profit-aware papers rarely explain individual predictions, and XAI papers rarely cost their recommendations. The framework proposed here sits deliberately at the intersection: Shapley attributions supply the reasons, and an intervention taxonomy with business ownership supplies the economically motivated response.

D. Research Gap

Three gaps emerge from the review. First, the accuracy-interpretability relationship in churn prediction is usually presented as a trade-off to be accepted rather than an interaction to be measured; studies seldom report how much discrimination is sacrificed, if any, when an interpretable pipeline is required, and seldom quantify explanation stability at all. Second, no widely reported study translates per-customer explanation outputs into a structured, auditable set of retention actions for non-technical stakeholders; the explanation is treated as the end of the analysis rather than the beginning of the decision. Third, cross-domain validation is rare, leaving open the question of whether churn drivers identified in one sector transfer to another [15]. The framework described in the following sections is constructed to address all three gaps directly: it measures accuracy and explanation quality jointly, it operationalises explanations through an intervention taxonomy and dashboard, and it validates across telecommunications, banking, and electronic commerce data.

III. PROPOSED FRAMEWORK

The framework is organised as a pipeline of five layers, illustrated in Fig. 1, each of which consumes the outputs of the previous one. The data layer acquires and prepares the three datasets described in Section IV. The feature layer organises predictors into behavioural, transactional, contractual, and engagement categories, giving every downstream explanation a vocabulary that business users already speak. The model layer trains the three classifiers and selects, per dataset, the model that offers the best joint profile of discrimination, calibration, and explanation quality. The explanation layer computes SHAP values for every scored customer and validates them against LIME on a fixed audit sample. The decision layer maps explanations onto retention interventions and presents the result through a dashboard designed for CRM practitioners.

Two design principles run through the pipeline. The first is that interpretability is a requirement, not a preference, so every modelling choice is assessed for its effect on the explanation layer as well as on predictive performance. The second is that every automated recommendation must remain open to human revision; the system proposes, and the retention team disposes.

These principles shape concrete choices reported below, including the retention of logistic regression as a permanent baseline and the decision to expose driver categories rather than raw feature names in the user interface.

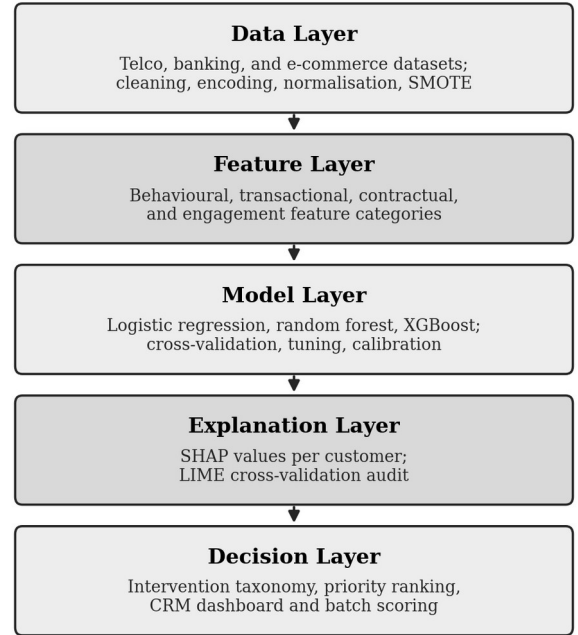


Fig. 1. System architecture of the proposed framework. Each layer consumes the outputs of the layer above it, ending in prioritised retention recommendations delivered through the dashboard.

The pipeline is also an accountability structure. Every trained model is stored in a versioned registry together with its training data snapshot, hyperparameters, evaluation report, and the explanation audit described in Section V, and every score served to the dashboard records the model version that produced it. This bookkeeping is unglamorous but essential: when a retention decision is later questioned, whether by a manager, an auditor, or the customer concerned, the framework can reproduce the prediction, the explanation, and the recommendation exactly as they stood at the time of the decision. The same records support the monitoring regime set out in Section VI, since drift in the served population can be measured against the stored training snapshot.

IV. METHODOLOGY

A. Datasets

Three publicly available, pre-labelled datasets were selected to represent the three sectors in which subscription churn is most studied. Their principal characteristics are summarised in Table I. The IBM Telco Customer Churn dataset is the most widely used benchmark in the field and records demographic, service-subscription, contract, and billing attributes for 7,043 customers of a fictional telecommunications operator, with a churn rate of approximately 26.5%. The Bank Customer Churn dataset covers 10,000 retail banking clients with attributes including credit score, account balance, product count, and estimated salary, with roughly 20.4% of clients marked as exited. The E-Commerce Customer Churn dataset comprises 5,630 records

from an online retail platform and is the most behavioural of the three, with fields such as time spent on the application, days since the last order, complaint history, and cashback received, and a churn rate near 16.8%.

The combination is deliberate. The telecommunications dataset anchors the study to the literature and enables comparison with published baselines; the banking dataset tests transfer to a domain governed by different contractual dynamics; the e-commerce dataset stresses the framework with behavioural, high-frequency features of the kind Alboukaey et al. found most predictive [12]. All three are anonymised, free of direct identifiers, and distributed for research use, which keeps the ethical footprint of the study small. Their churn rates, ranging from roughly 0.17 to 0.27 of records, are representative of the class imbalance that retention modelling must handle in practice.

TABLE I. CHARACTERISTICS OF THE SELECTED DATASETS

Dataset	Domain	Records	Features	Churn Rate
IBM Telco Customer Churn (Kaggle)	Telecommunications	7,043	21	26.5%
Bank Customer Churn (Kaggle)	Retail banking	10,000	13	20.4%
E-Commerce Customer Churn (Kaggle)	Electronic commerce	5,630	20	16.8%

B. Data Preparation

Preparation follows an identical protocol on each dataset so that cross-domain comparisons are not confounded by differing preprocessing. Records with duplicate identifiers are removed, and the small number of fields with missing values, such as the total charges attribute in the Telco data, are imputed with median values for continuous variables and a distinct category for categorical ones. Categorical predictors are one-hot encoded. Continuous predictors are scaled to the unit interval with min-max normalisation, which logistic regression requires for stable coefficient estimation and which does not harm tree-based models. Each dataset is then split into training and test partitions in an 80/20 ratio, stratified on the churn label.

Class imbalance is addressed with SMOTE [14], applied under two constraints that the literature identifies as necessary for honest evaluation [13]. Synthetic examples are generated only from training data, never from test data, and when cross-validation is used the oversampling is performed inside each fold rather than before the split, so that no synthetic point derived from a validation instance can leak into the corresponding training set. Test partitions retain their natural class distribution throughout, since performance claims are only credible against the imbalance the model will face in production.

C. Feature Engineering

Predictors are organised into four categories, which serve both as an engineering checklist and, later, as the vocabulary of the explanation layer.

- Behavioural features: login or session frequency, product usage rate, breadth of feature adoption, and

support ticket frequency, computed over recent windows where timestamps permit.

- Transactional features: average monthly spend, payment method, billing cycle, cashback or discount received, and outstanding balance.
- Contractual features: contract type, tenure, renewal history, and the presence of value-added services such as online security or technical support.
- Engagement features: complaint frequency, satisfaction proxies where available, and response to past promotional contact.

Derived features follow the evidence on temporal dynamics [12]: recency variables such as days since last order, and ratio variables such as charge per unit of tenure, are added where the raw fields allow. Feature selection is kept conservative. Predictors with near-zero variance are dropped, and one member of any pair with absolute correlation above 0.9 is removed, since strongly correlated features both destabilise logistic regression coefficients and dilute Shapley attributions across duplicates [18]. No aggressive automated selection is applied, because the explanation layer, not a selection algorithm, is where the framework interrogates feature relevance.

D. Model Selection and Justification

Three classifiers are trained on every dataset. They were chosen to form a ladder of increasing representational power with decreasing native transparency, which is precisely the trade-off the study is designed to measure.

Logistic regression is the baseline. It models the churn probability as shown in (1), where x is the feature vector, β the coefficient vector, and β subscript zero the intercept.

$$p(x) = 1 / (1 + \exp(-(\beta_0 + \beta^T x))) \quad (1)$$

Its advantages are exact interpretability, since each coefficient is a log-odds contribution, together with fast training, stable behaviour on modest samples, and well-calibrated probabilities. Its limitation is the linearity assumption: interactions and threshold effects, which are common in churn data, must be hand-crafted or are missed. It is retained not despite this simplicity but because of it, as the reference against which the cost of opacity is priced.

Random forest aggregates several hundred decorrelated decision trees grown on bootstrap samples with random feature subsets [22]. It captures non-linearities and interactions without manual engineering, is robust to outliers and to the mixed feature types typical of CRM data, and its bagged structure resists overfitting. Its drawbacks are the loss of direct interpretability, a tendency to dilute importance across correlated features, and probability estimates that often require recalibration.

XGBoost implements regularised gradient boosting, building trees sequentially so that each corrects the residual errors of its predecessors, with explicit penalties on tree complexity [23]. It is the strongest performer on tabular churn

benchmarks [4], [5], [15] and handles imbalance through instance weighting in addition to resampling. Its costs are sensitivity to hyperparameters, a greater risk of overfitting small datasets than bagging, and the same native opacity as any large ensemble.

Deep neural architectures were considered and set aside. The systematic evidence indicates that on tabular data of the size used here they seldom beat tuned gradient boosting, while demanding more data, more tuning, and heavier explanation machinery [15]. All three retained models are compatible with exact or efficient Shapley computation, logistic regression trivially and the two ensembles through TreeExplainer [18], which keeps the explanation layer computationally honest.

E. Training Protocol and Hyperparameter Optimisation

Model development uses stratified ten-fold cross-validation on the training partition, with SMOTE applied inside each fold as described above. Logistic regression is tuned over the inverse regularisation strength under an L2 penalty. Random forest is tuned by grid search over the number of estimators, maximum depth, minimum samples per split, and maximum features per split. XGBoost, whose parameter space is too large for exhaustive search, is tuned by Bayesian optimisation with the Optuna framework over learning rate, maximum depth, subsampling ratios, and the number of boosting rounds, with early stopping on a validation fold [24]. After selection, probability outputs of the ensemble models are calibrated on held-out folds, and every model is refitted on the full training partition before a single evaluation on the untouched test set. The implementation uses the standard Python scientific stack, and all random seeds are fixed and reported so that results can be reproduced exactly.

Classification requires an operating threshold as well as a probability, and the choice is a business decision disguised as a technical one. Rather than defaulting to 0.5, the threshold is selected on validation folds to maximise F1-score subject to a minimum precision agreed with the retention budget holder, since precision determines how much of the campaign budget is spent on customers who were never going to leave. Where campaign costs and customer values are available, the threshold is instead set to maximise expected campaign profit, following the logic of the expected maximum profit criterion [19]. The selected threshold is reported with every model so that published performance figures correspond to the configuration the decision layer actually uses.

F. Evaluation Metrics

Accuracy alone is uninformative under imbalance: a classifier that never predicts churn is 83% accurate on a dataset with a churn rate of 0.17. The evaluation therefore reports a suite of complementary metrics. Precision, defined in (2), measures the proportion of predicted churners who actually churn, and governs how much retention budget is wasted on customers who were never leaving. Recall, defined in (3), measures the proportion of actual churners the model catches, and governs how much revenue walks out unnoticed. The F1-score, defined in (4), is their harmonic mean and is reported as a single-number balance of the two.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad (2)$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad (3)$$

$$F_1 = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall}) \quad (4)$$

Here TP, FP, and FN denote true positives, false positives, and false negatives respectively. Threshold-free discrimination is reported as the area under the receiver operating characteristic curve (ROC-AUC). Because ROC-AUC can flatter models under heavy imbalance, the Matthews correlation coefficient (MCC) is reported alongside it, following the evidence of Chicco and Jurman that MCC is the more conservative summary of a confusion matrix [25]. Probability calibration is assessed with the Brier score, since the decision layer consumes probabilities rather than labels and a miscalibrated 0.8 is operationally misleading. Finally, because the point of the system is economic, the evaluation includes a profit-oriented reading in the spirit of the expected maximum profit criterion, estimating campaign value at the operating threshold chosen by the decision layer [19]. Explanation quality is assessed in Section V alongside these predictive metrics.

V. EXPLAINABILITY AND DECISION SUPPORT

A. Interpreting Shapley Explanations

SHAP expresses every prediction as an additive attribution, as shown in (5), where g is the explanation model, z' the simplified feature presence vector, M the number of features, ϕ subscript zero the base value equal to the mean model output, and each ϕ subscript j the Shapley value of feature j [8].

$$g(z') = \phi_0 + \sum_j \phi_j z'_j \quad (5)$$

Three views of these attributions are produced for the tree ensembles with TreeExplainer [18]. Global feature importance, computed as mean absolute Shapley value, tells the business which drivers dominate churn across the whole customer base and is the natural input to strategic decisions such as pricing review. Beeswarm summary plots add direction, showing for instance that short tenure raises churn risk while long tenure suppresses it. Per-customer waterfall charts decompose one individual's risk estimate into named contributions, and it is this local view that the decision layer consumes. Interpretation is disciplined by two rules. Attributions are read as descriptions of model behaviour, not as causal claims about customers, a distinction Rudin's critique makes unavoidable [6]; and any driver whose direction contradicts domain knowledge is escalated for data-quality review before the model is trusted.

B. Cross-Validation with Local Surrogate Models

Because post-hoc explanations can be fragile, SHAP outputs are audited against LIME on a fixed sample of 100 test-set predictions per dataset, stratified across the risk spectrum [9]. For each sampled customer the top five features by absolute attribution are extracted from both methods and compared by rank agreement. High agreement raises confidence that the explanation reflects the model rather than the idiosyncrasies of one explainer; systematic disagreement

localises the features whose attributions should not be shown to business users without caveats. This audit, together with the stability of repeated LIME runs, constitutes the explanation-quality component of the evaluation protocol.

C. From Explanations to Retention Actions

The step that distinguishes this framework is the mapping from explanation to intervention. SHAP drivers are first grouped into the four feature categories of Section IV, then matched to a retention intervention taxonomy developed with reference to the retention-marketing literature [3], [2]. Table II illustrates the mapping. Each rule carries a priority derived from the customer's churn probability band and estimated value, so that the output of the system is a ranked work queue rather than an undifferentiated list. A customer whose risk is driven by month-to-month contract status and high monthly charge, for example, is routed to the pricing team with a contract-upgrade offer, whereas one driven by rising complaint frequency is routed to service recovery. The mapping is stored as an editable rule table, not code, so that retention managers can adjust it as offers change, and every recommendation records the drivers that produced it, which keeps the decision trail auditable.

TABLE II. ILLUSTRATIVE MAPPING FROM CHURN DRIVERS TO RETENTION INTERVENTIONS

SHAP Driver (Category)	Recommended Intervention	Business Owner
Month-to-month contract (contractual)	Offer fixed-term upgrade with introductory discount	Pricing team
High monthly charge (transactional)	Tariff review; targeted loyalty pricing	Pricing team
Short tenure (contractual)	Structured onboarding contact in first 90 days	Customer success
Rising complaints (engagement)	Priority service recovery call	Service recovery
Falling usage (behavioural)	Re-engagement campaign; feature education	Marketing
No value-added services (contractual)	Bundle offer for security or support add-ons	Sales

D. Decision-Support Dashboard

The dashboard is implemented in Streamlit, chosen over a Flask and React stack because it keeps the interface in the same Python environment as the models and reduces maintenance to a single codebase. A CRM user uploads a customer list as a comma-separated file and receives, for each customer, the churn probability expressed as a banded score, the three strongest churn drivers stated in category language rather than raw feature names, and the recommended intervention with its priority. Batch summaries show the distribution of risk across the uploaded base and the aggregate driver profile, which supports campaign-level planning as well as case-level action. Nothing in the interface requires statistical literacy: the design assumes a user who knows the customers and the offers, not the models. An export function returns the scored list for ingestion into existing CRM systems, and every screen carries the model version and scoring date so that decisions can be traced to the model that informed them.

A short walkthrough makes the intended use concrete. A retention analyst preparing the month's campaign uploads the active customer base on a Monday morning. The dashboard returns 412 customers in the high-risk band. Sorting by priority, the analyst sees that the largest driver group is contractual, month-to-month customers approaching a price anniversary, and routes the corresponding queue to the pricing team's contract-upgrade offer. A smaller group driven by complaint frequency goes to service recovery the same day, because complaint-driven risk decays fastest. Two recommendations look wrong to the analyst, who knows that the customers involved recently renewed; both are overridden, and the overrides enter the log that feeds the next model review. Nothing in the session required the analyst to interpret a probability curve, yet every action taken is traceable to a quantified, explained risk estimate.

VI. ETHICAL AND PRACTICAL CONSIDERATIONS

A system that uses personal data to decide who receives retention offers carries obligations beyond predictive performance. Privacy comes first. The study itself uses anonymised public datasets, but the framework is designed for production use, so it assumes pseudonymised identifiers, data minimisation in the feature set, and role-based access to the dashboard. Under the General Data Protection Regulation (GDPR), customers have a qualified right to meaningful information about automated decisions that significantly affect them, and the per-customer explanations the framework already produces are the natural mechanism for honouring it [7].

Purpose limitation deserves separate mention. Data collected to operate a service is being reused here to predict and influence customer behaviour, and that reuse must be covered by the lawful basis under which the data were gathered and disclosed in the privacy notice customers actually receive. The feature set is reviewed against this test before training: variables whose collection purpose is remote from retention, or whose presence a reasonable customer would find surprising, are excluded even when they carry predictive signal. Retention periods apply to the modelling data as they do to the operational data from which it derives, and scored lists exported from the dashboard inherit the access controls of the CRM system they enter.

Fairness requires equal attention. Churn models trained on historical data can encode historical inequities, and features such as geography or salary can proxy for protected attributes [26]. The framework therefore includes a fairness audit: performance and error rates are compared across available demographic segments, Shapley attributions are inspected for proxy behaviour, and any feature functioning as a proxy without independent business justification is removed. There is also a distributional question specific to retention: if offers flow only to customers the model flags, systematically unflagged groups quietly receive worse treatment. Publishing the intervention taxonomy and keeping a human review step in the loop are partial but practical safeguards. Finally, the system is advisory by design. Retention staff can override any recommendation, overrides are logged, and the log is reviewed

both to improve the model and to detect automation bias, the tendency to defer to the system precisely because it is a system.

Practical constraints are acknowledged with the same directness. Customer behaviour drifts, so deployed models require scheduled retraining and monitoring of both performance and calibration. Explanation computation adds latency and cost at scoring time, which the design contains by using TreeExplainer and by precomputing explanations during batch scoring. The rule table that maps drivers to interventions embodies commercial judgement that will differ between firms, and is exposed for exactly that reason.

VII. LIMITATIONS AND FUTURE WORK

The proposal has limits that should be stated plainly. The three datasets, though drawn from different sectors, are public benchmarks of modest size, and the Telco data in particular describes a fictional operator; conclusions about cross-domain transfer therefore need confirmation on proprietary data before they can guide practice. SMOTE generates synthetic minority examples by interpolation, and although it is confined to training folds, it can still smooth over rare but genuine churn patterns [13]. Shapley attributions describe the model rather than the customer, so the retention actions they trigger are motivated correlations, not established causes; the framework flags customers and reasons, but it cannot promise that treating the reason will avert the churn. The dashboard, finally, is designed against personas rather than validated with users, and the audit of explanations relies on agreement between two post-hoc methods, which is evidence of robustness rather than proof of fidelity.

The intervention taxonomy carries its own untested assumptions. It encodes a judgement that a given driver is best answered by a given offer, and that judgement rests on the retention-marketing literature and on face validity rather than on experimental evidence from the firms that would use it. Different sectors price interventions differently, and a mapping that is profitable for a telecommunications operator may be loss-making for a bank. The taxonomy is editable for precisely this reason, but editability transfers the burden rather than discharging it, and the framework's end-to-end value therefore remains a hypothesis until it is measured against retention outcomes in the field.

Each limitation points to further work. The natural methodological extension is uplift modelling, which estimates the effect of an intervention on each customer and would convert the decision layer from risk-based to effect-based targeting [21]. Profit-driven training objectives could replace post-hoc profit evaluation, aligning the models themselves with the economics of retention [20], [19]. A user study with practising CRM staff would test whether the explanations change decisions as intended. On the modelling side, sequence models over event streams merit examination where fine-grained behavioural logs exist [12], and on the deployment side, a longitudinal field trial measuring realised retention against a control group is the standard the framework should ultimately meet.

VIII. CONCLUSION

This paper has argued that the useful frontier in churn prediction is no longer accuracy but the distance between a probability and a decision. It has proposed a framework that closes that distance by pairing three classifiers of graded complexity with exact Shapley explanations, auditing those explanations against an independent method, and translating them through an explicit taxonomy into prioritised retention actions delivered in a dashboard built for non-technical staff. Evaluation is designed to match: discrimination, calibration, and explanation quality are measured together, across three sectors rather than one, with ethical safeguards treated as design requirements. If churn research is to influence retention practice, its outputs must arrive in the language of the people who act on them; the framework presented here is a concrete attempt to make that translation systematic, measurable, and accountable.

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REFERENCES

- [1] J. Hadden, A. Tiwari, R. Roy, and D. Ruta, "Computer assisted customer churn management: state-of-the-art and future trends," *Comput. Oper. Res.*, vol. 34, no. 10, pp. 2902–2917, 2007.
- [2] W. Verbeke, K. Dejaeger, D. Martens, J. Hur, and B. Baesens, "New insights into churn prediction in the telecommunication sector: a profit driven data mining approach," *Eur. J. Oper. Res.*, vol. 218, no. 1, pp. 211–229, 2012.
- [3] A. Lemmens and S. Gupta, "Managing churn to maximize profits," *Mark. Sci.*, vol. 39, no. 5, pp. 956–973, 2020.
- [4] T. Vafeiadis, K. I. Diamantaras, G. Sarigiannidis, and K. C. Chatzisavvas, "A comparison of machine learning techniques for customer churn prediction," *Simul. Model. Pract. Theory*, vol. 55, pp. 1–9, 2015.
- [5] A. K. Ahmad, A. Jafar, and K. Aljoumaa, "Customer churn prediction in telecom using machine learning in big data platform," *J. Big Data*, vol. 6, no. 1, art. 28, 2019.
- [6] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nat. Mach. Intell.*, vol. 1, no. 5, pp. 206–215, 2019.
- [7] B. Goodman and S. Flaxman, "European Union regulations on algorithmic decision-making and a right to explanation," *AI Mag.*, vol. 38, no. 3, pp. 50–57, 2017.
- [8] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems 30*, Long Beach, CA, USA, 2017, pp. 4765–4774.
- [9] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you? Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, San Francisco, CA, USA, 2016, pp. 1135–1144.
- [10] A. El Attar and M. El-Hajj, "Explainable AI-driven customer churn prediction: a multi-model ensemble approach with SHAP-based feature analysis," *Front. Artif. Intell.*, art. 1748799, 2026.
- [11] A. De Caigny, K. Coussement, and K. W. De Bock, "A new hybrid classification algorithm for customer churn prediction based on logistic regression and decision trees," *Eur. J. Oper. Res.*, vol. 269, no. 2, pp. 760–772, 2018.
- [12] N. Alboukaey, A. Joukhadar, and N. Ghneim, "Dynamic behavior based churn prediction in mobile telecom," *Expert Syst. Appl.*, vol. 162, art. 113779, 2020.

- [13] J. Burez and D. Van den Poel, "Handling class imbalance in customer churn prediction," *Expert Syst. Appl.*, vol. 36, no. 3, pp. 4626–4636, 2009.
- [14] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, 2002.
- [15] M. Imani, M. Joudaki, A. Beikmohammadi, and H. R. Arabnia, "Customer churn prediction: a systematic review of recent advances, trends, and challenges in machine learning and deep learning," *Mach. Learn. Knowl. Extr.*, vol. 7, no. 3, art. 105, 2025.
- [16] A. Adadi and M. Berrada, "Peeking inside the black-box: a survey on explainable artificial intelligence (XAI)," *IEEE Access*, vol. 6, pp. 52138–52160, 2018.
- [17] R. Guidotti, A. Monreale, S. Ruggieri, F. Turini, F. Giannotti, and D. Pedreschi, "A survey of methods for explaining black box models," *ACM Comput. Surv.*, vol. 51, no. 5, art. 93, 2018.
- [18] S. M. Lundberg, G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, and S.-I. Lee, "From local explanations to global understanding with explainable AI for trees," *Nat. Mach. Intell.*, vol. 2, no. 1, pp. 56–67, 2020.
- [19] T. Verbraken, W. Verbeke, and B. Baesens, "A novel profit maximizing metric for measuring classification performance of customer churn prediction models," *IEEE Trans. Knowl. Data Eng.*, vol. 25, no. 5, pp. 961–973, 2013.
- [20] S. Höppner, E. Stripling, B. Baesens, S. vanden Broucke, and T. Verdonck, "Profit driven decision trees for churn prediction," *Eur. J. Oper. Res.*, vol. 284, no. 3, pp. 920–933, 2020.
- [21] F. Devriendt, D. Moldovan, and W. Verbeke, "A literature survey and experimental evaluation of the state-of-the-art in uplift modeling: a stepping stone toward the development of prescriptive analytics," *Big Data*, vol. 6, no. 1, pp. 13–41, 2018.
- [22] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.
- [23] T. Chen and C. Guestrin, "XGBoost: a scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, San Francisco, CA, USA, 2016, pp. 785–794.
- [24] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: a next-generation hyperparameter optimization framework," in *Proc. 25th ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, Anchorage, AK, USA, 2019, pp. 2623–2631.
- [25] D. Chicco and G. Jurman, "The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation," *BMC Genomics*, vol. 21, art. 6, 2020.
- [26] N. Mehrabi, F. Morstatter, N. Saxena, K. Lerman, and A. Galstyan, "A survey on bias and fairness in machine learning," *ACM Comput. Surv.*, vol. 54, no. 6, art. 115, 2021.